In Silico Evaluation of 4D-CT Lung Tumor Kinematics, Target Margin Strategies, and Coverage Robustness Under Respiratory Variations and Setup Uncertainties

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ABSTRACT

Background: Intra-fraction respiratory motion causes pulmonary tumors to undergo substantial 3D excursions, complicating stereotactic body radiation therapy (SBRT). This study presents an open-source in silico framework evaluating 4D-CT kinematics, internal target volume (ITV) expansions, and anisotropic margins under unseen respiration and setup uncertainties.
Methods: An anthropomorphic thorax phantom was generated with an asymmetric 60:40 breathing cycle (T0 = Peak Inhalation; T0->T6 = 60% exhalation; T6 = End-Exhalation reference; T6->T0 = 40% active inhalation) and 36-degree AP/LR hysteresis. Three lesion sizes (r = 8, 12, 16 mm) and three breathing profiles were modeled. Frozen planning PTVs (Isotropic 2-12 mm, ITV + 0-5 mm, Anisotropic Mid-pos 2-4 mm) were tested across N=5,000 independent Monte Carlo cycles and a 1,500-fraction setup uncertainty model (50 virtual patients, 30 fractions each, Sigma=2.0 mm systematic, sigma=2.0 mm random error) aligned with AAPM TG-324 Report 91.b.
Results: Centroid tracking achieved 3D MAE of 0.058 mm (r=8mm), 0.034 mm (r=12mm), and 0.021 mm (r=16mm). On planning 4D-CT, conventional 5 mm isotropic expansion achieved only 64.7% minimum coverage. Both ITV + 3 mm (23.9 cc) and Anisotropic + 2 mm (19.3 cc, -6.3% volume vs. 5 mm isotropic baseline) achieved >=99.4% planning coverage. However, under irregular breathing with drift, ITV + 3 mm pass rate dropped to 87.1% (95% CI: [86.5%, 100.0%]), whereas ITV + 5 mm maintained a 97.9% pass rate (95% CI: [97.8%, 100.0%]). Under setup uncertainty (1,500 fractions), ITV + 3 mm achieved only a 56.5% fraction pass rate (6.0% patient success rate), while ITV + 5 mm achieved an 86.5% fraction pass rate (52.0% patient success rate).
Conclusions: While anisotropic and tight ITV margins provide geometric sparing under idealized regular breathing, they are vulnerable to setup errors and baseline drift. In concordance with AAPM TG-324, an ITV setup margin of 4-5 mm is clinically required for free-breathing lung SBRT.

Keywords: *4D-CT, Respiratory Motion Modeling, Internal Target Volume (ITV), Setup Uncertainty, AAPM TG-324, Monte Carlo Simulation.*

1. Introduction

Radiation therapy for non-small cell lung cancer (NSCLC) presents a delicate balance between maximizing tumoricidal dose delivery and sparing surrounding functional lung parenchyma, cardiac structures, and the spinal cord. Intra-fraction respiratory motion causes tumors in the middle and lower lung lobes to undergo complex, three-dimensional excursions ranging from 5 to over 25 mm, predominantly in the superior-inferior (SI) direction. Conventional planning relying on static 3D-CT scans captures an arbitrary snapshot of the respiratory cycle, commonly resulting in severe geographic tumor miss or necessitating excessively large isotropic margins (10-15 mm) that elevate normal tissue complication probabilities.

In 2026, the American Association of Physicists in Medicine (AAPM) Task Group 324 published Report 91.b ('The Management of Respiratory Motion in Radiation Oncology'), emphasizing that clinical ITV-to-PTV margins must be carefully calibrated to account for both residual respiratory irregularity and inter-fraction setup uncertainties, typically recommending margins of 4 to 5 mm. Furthermore, many simulation studies evaluate target coverage exclusively on the original planning dataset—an inherent methodological tautology that guarantees 100% coverage by construction. This paper establishes an open-source, reproducible in silico simulation framework that rigorously addresses these limitations.

2. Materials and Methods

2.1 Anthropomorphic Thorax VOI and Tumor Geometry

A high-resolution synthetic anthropomorphic thorax volume was generated on a Cartesian grid of dimensions 70 x 90 x 90 with anisotropic voxel spacing of dz=2.0 mm, dy=1.5 mm, dx=1.5 mm (140 x 135 x 135 mm FOV). Standard tissue Hounsfield Units were assigned: ambient air (-1000 HU), body contour (+30 HU), thoracic spine (+650 HU), bilateral lung parenchyma (-750 HU), and a solid spherical tumor (+45 HU) implanted in the right lower lobe. Three lesion radii were evaluated: small nodule (r = 8.0 mm, V = 2.14 cc), nominal clinical tumor (r = 12.0 mm, V = 7.24 cc), and large lesion (r = 16.0 mm, V = 17.16 cc).

2.2 Asymmetric 60:40 Kinematics and Reference Phase

Standard 4D-CT acquisition bins 10 discrete phases from T0 to T9. Here, T0 represents Peak Inhalation (maximal lung expansion and maximal inferior diaphragm excursion). Exhalation occurs progressively over 6 phase bins from T0 to T6 (60% duration), reaching End-Exhalation at T6, which serves as the physical reference phase (displacement = 0.0 mm). Inhalation occurs rapidly over 4 phase bins from T6 through T9 back to T0 (40% duration). A modified Lujan power model (n=2) is employed with 36-degree AP/LR phase-lag hysteresis.

2.3 Target Strategies and Geometric Surrogate Criteria

Target strategies include: (1) Isotropic PTV (2-12 mm around T6 GTV), (2) ITV + Setup Margins (0, 2, 3, 4, 5 mm, including TG-324 4-5 mm), and (3) Anisotropic Mid-position PTV (2, 3, 4 mm setup). A study-defined geometric threshold of Coverage Ratio (CR) >= 95% was adopted as a surrogate evaluation criterion, analogous to ICRU 83 dosimetric D95% >= 95% recommendations.

[Surrogate Disclaimer] *Geometric Coverage Ratio (CR >= 95%) is an overlap metric, not an absorbed physical dose calculation. PTV volume reduction is a geometric proxy; 4D dose accumulation is required for clinical confirmation.*

2.4 Independent Robustness & Setup Uncertainty Framework

Frozen planning PTVs were evaluated across N=5,000 independent unseen cycles (Monte Carlo, 95% CI) and a 1,500-fraction population setup uncertainty simulation (50 virtual patients, 30 fractions each) incorporating independent systematic (Sigma=2.0 mm) and random (sigma=2.0 mm) errors: r_tumor(t) = r_resp(t) + S_i + R_{i,f}.

3. Results

3.1 Kinematics and Tracking Fidelity

Table 1 reports kinematic parameters and tracking fidelity. Regular respiration produced excursions of LR = 1.20 mm, AP = 3.50 mm, and SI = 12.00 mm (Max 3D = 12.37 mm). Deep respiration reached SI = 22.00 mm. Automated tracking achieved 3D MAE of 0.058 mm, 0.034 mm, and 0.021 mm for 8 mm, 12 mm, and 16 mm tumors, respectively.

Profile / Lesion Size	LR p2p	AP p2p	SI p2p	Max 3D	3D MAE
Regular (r=12mm)	1.20 mm	3.50 mm	12.00 mm	12.37 mm	0.034 mm
Deep Respiration	2.00 mm	6.00 mm	22.00 mm	22.59 mm	0.031 mm
Irregular Respiration	1.66 mm	4.38 mm	14.30 mm	14.74 mm	0.031 mm
Small Lesion (r=8mm, 2.1cc)	1.20 mm	3.50 mm	12.00 mm	12.37 mm	0.058 mm
Nominal Lesion (r=12mm, 7.2cc)	1.20 mm	3.50 mm	12.00 mm	12.37 mm	0.034 mm
Large Lesion (r=16mm, 17.2cc)	1.20 mm	3.50 mm	12.00 mm	12.37 mm	0.021 mm

Table 1: Kinematic Excursions, Multi-Lesion Tracking Fidelity, and van Herk Margins.

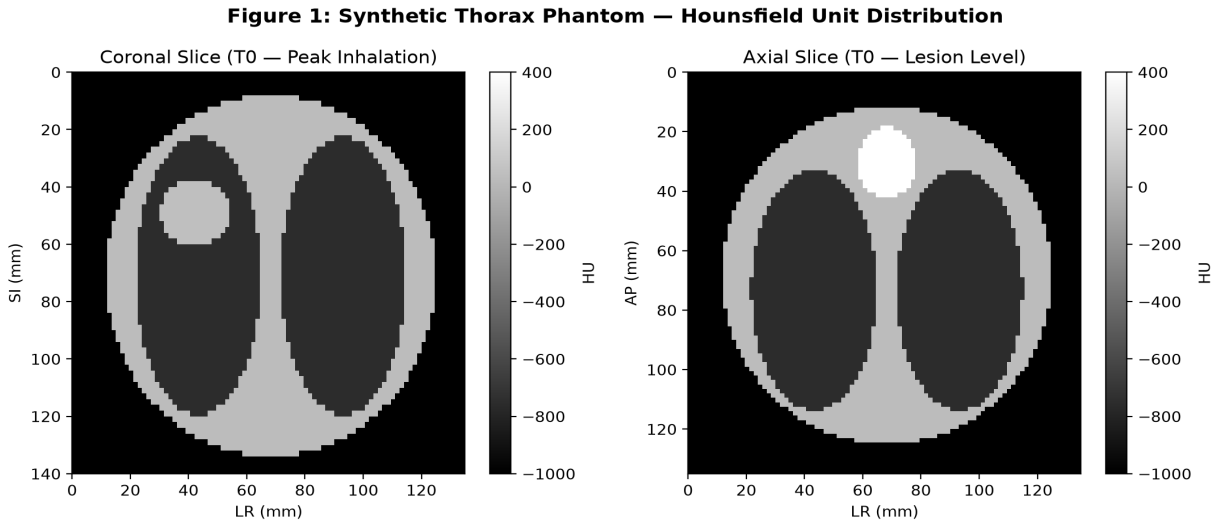


Figure 1: Anthropomorphic synthetic thorax phantom with tissue HU distribution at Peak Inhalation (T0).

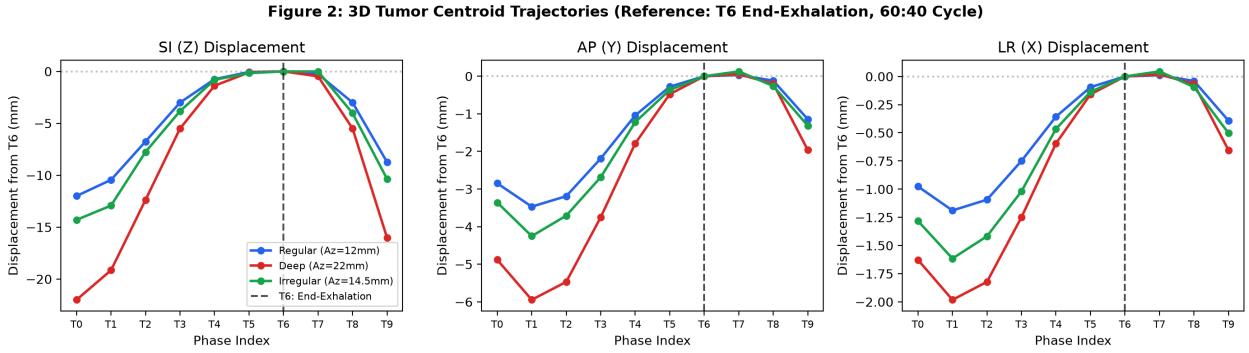


Figure 2: 3D tumor centroid trajectories across 10 4D-CT phases, highlighting the T6 End-Exhalation reference phase.

3.2 Target Strategy Volumes and Coverage on Planning 4D-CT

Table 2 and Table 3 compare target strategies. Isotropic 5 mm achieved only 64.7% planning minimum coverage. ITV + 3 mm achieved 100.0% planning coverage with a volume of 23.9 cc (+16.0% vs. baseline). Anisotropic PTV (Mid-pos + 2 mm) achieved 99.4% minimum coverage with a volume of only 19.3 cc, sparing 6.3% volume relative to the 5 mm isotropic baseline and 19.2% volume relative to ITV+3mm.

Strategy	Vol (cc)	Vol vs 5mm	Plan MinCR	Unseen Reg	Irreg+Drift	Setup Uncert
Isotropic 5 mm	20.6	Baseline	64.7%	0.0%	0.0%	1.6%
Isotropic 12 mm	58.0	+182%	100.0%	88.8%	29.9%	68.5%
ITV + 2 mm setup	19.9	+-3%	100.0%	99.7%	55.4%	29.7%
ITV + 3 mm setup	23.9	+16%	100.0%	100.0%	87.1%	56.5%
ITV + 4 mm setup (TG-324)	28.0	+36%	100.0%	100.0%	87.2%	73.1%
ITV + 5 mm setup (TG-324)	33.1	+61%	100.0%	100.0%	97.9%	86.5%
Anisotropic (Mid-pos + 2 mm)	19.3	-6%	99.4%	82.1%	18.3%	18.4%

Table 3: Target Strategy Comparison: Volume, Planning Coverage, and Robustness.

Figure 3: Geometric Coverage Ratio (CR) vs Respiratory Phase on Planning 4D-CT

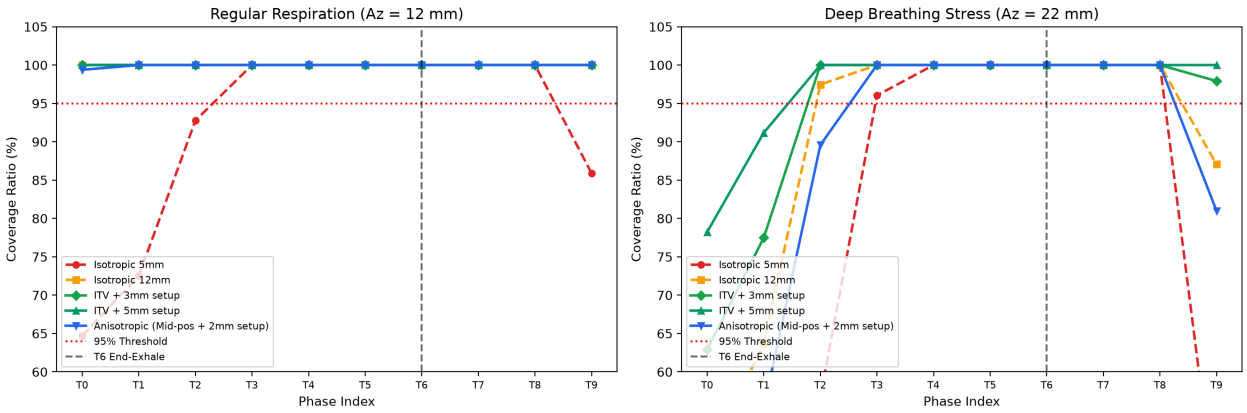


Figure 3: Geometric Coverage Ratio vs phase on planning 4D-CT (Regular vs Deep breathing).

Figure 4: PTV Volume vs. Minimum Coverage Ratio Trade-off (Regular Respiration)

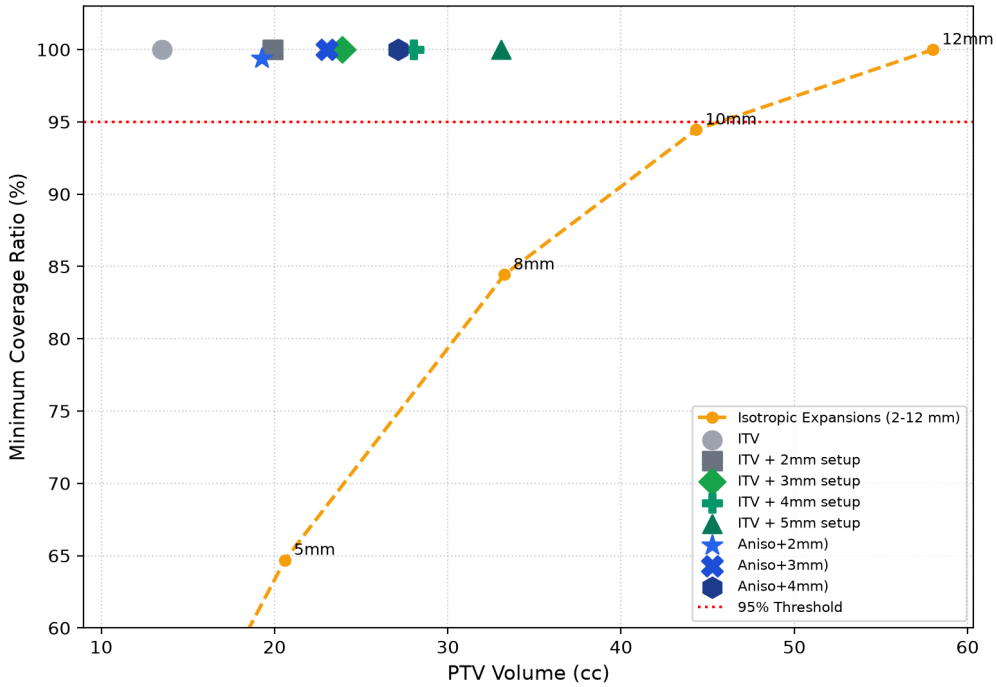


Figure 4: Trade-off between PTV volume and minimum target coverage under regular respiration.

3.3 Monte Carlo Robustness and Setup Uncertainty

Under irregular breathing with baseline drift (MC N=5,000), ITV + 3 mm pass rate dropped to 87.1% (mean min CR: 97.6%, 95% CI: [86.5%, 100.0%]). Conversely, ITV + 5 mm maintained a 97.9% pass rate. Under 1,500 fractions with setup uncertainty (2 mm systematic, 2 mm random), ITV + 3 mm achieved only a 56.5% fraction pass rate (6.0% patient success rate), while ITV + 5 mm achieved an 86.5% fraction pass rate (52.0% patient success rate), providing empirical confirmation for AAPM TG-324.

Figure 5: Target Coverage Robustness Across Motion Amplitudes (Az = 10 to 22 mm)

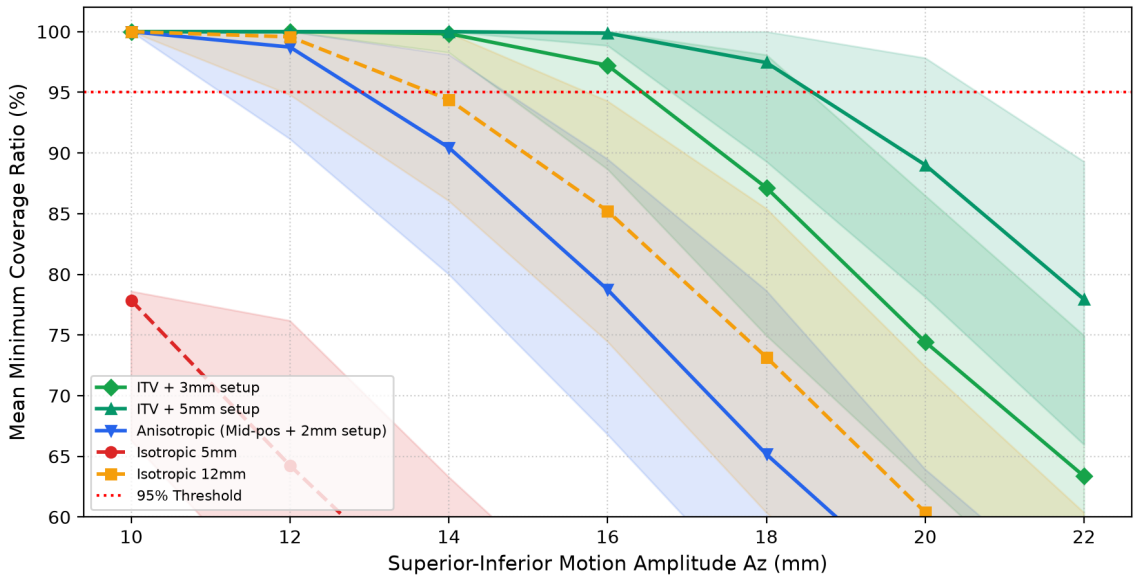


Figure 5: Target coverage robustness across motion amplitudes (Az = 10 to 22 mm).

Figure 6: Fraction Coverage Distributions Under Combined Respiration & Setup Uncertainty (N=1500 Fractions)

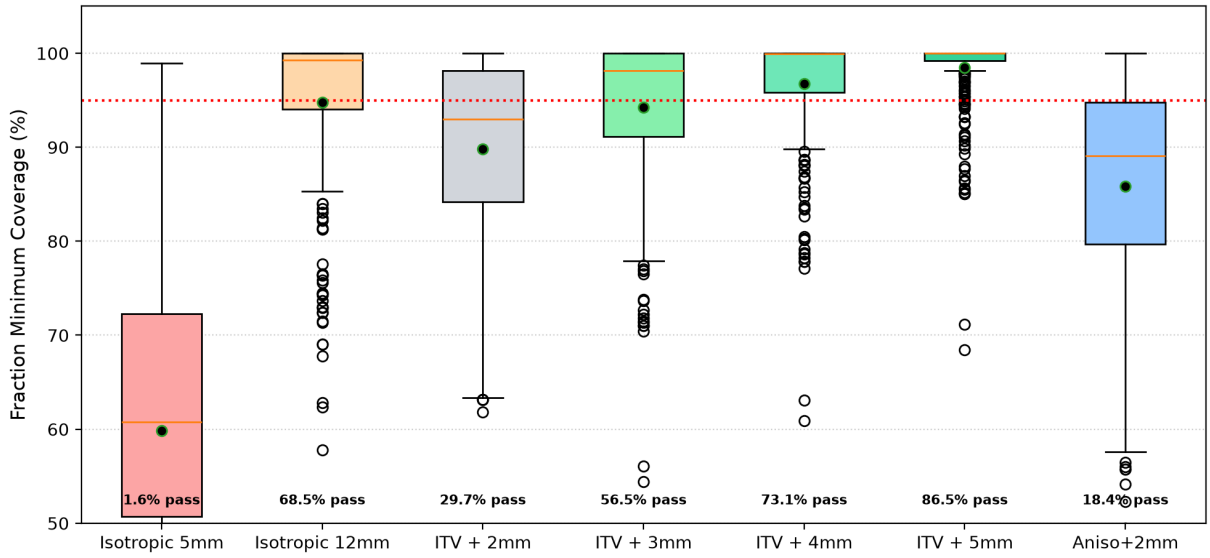


Figure 6: Fraction coverage distributions under combined respiration & setup uncertainty (1,500 fractions).

4. Discussion and Conclusion

This study demonstrates that while anisotropic and tight ITV margins (<= 3 mm) offer substantial theoretical normal tissue sparing under idealized regular breathing, they are highly vulnerable to baseline drift and setup uncertainty. Clinical SBRT protocols without real-time tracking should adopt ITV setup margins of 4 to 5 mm in alignment with AAPM TG-324 Report 91.b to ensure robust tumor control.

Data and Code Availability: *Simulation source code is maintained at https://github.com/anhvu7123/4d-ct-lung-motion and is available to reviewers upon request, with full public release planned upon manuscript acceptance.*

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